

PROGRAM:

```
#include <stdio.h>

#include <stdlib.h>

#define TABLE_SIZE 3

struct node {

    int data;

    struct node *next;

};

struct node *head[TABLE_SIZE] = {NULL};

void insert(int val) {

    int i = val % TABLE_SIZE;

    struct node *newnode = (struct node *)malloc(sizeof(struct node));

    if (newnode == NULL) {

        perror("Memory allocation failed");

        exit(EXIT_FAILURE);

    }

    newnode->data = val;

    newnode->next = NULL;

    if (head[i] == NULL) {

        head[i] = newnode;

    }

    else {

        struct node *c = head[i];

        while (c->next != NULL) {

            c = c->next;

        }

        c->next = newnode;

    }

}
```

```

}

void display() {

    for (int i = 0; i < TABLE_SIZE; i++) {

        printf("Entries at index %d: ", i);

        struct node *temp = head[i];

        if (temp == NULL) {

            printf("No Hash Entry\n");

        }

        else {

            while (temp != NULL) {

                printf("%d -> ", temp->data);

                temp = temp->next;

            }

            printf("NULL\n");

        }

    }

}

int main() {

    int opt, val;

    while (1) {

        printf("\nPress 1. Insert\t 2. Display \t3. Exit \n");

        scanf("%d", &opt);

        switch (opt) {

            case 1:

                printf("\nEnter a value to insert into hash table: ");

                scanf("%d", &val);

                insert(val);

                break;

```

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        case 2:
            display();
            break;
        case 3:
            exit(0);
        default:
            printf("\nInvalid option\n");
    }
}

return 0;
}

```

OUTPUT:

Press 1. Insert 2. Display 3. Exit

1

Enter a value to insert into hash table: 12

Press 1. Insert 2. Display 3. Exit

1

Enter a value to insert into hash table: 7

Press 1. Insert 2. Display 3. Exit

1

Enter a value to insert into hash table: 3

Press 1. Insert 2. Display 3. Exit

2

Entries at index 0: No Hash

Entry Entries at index 1: 12 -> NULL

Entries at index 2: 3 -> 7 -> NULL

Press 1. Insert 2. Display 3. Exit

1

Enter a value to insert into hash table: 15

Press 1. Insert 2. Display 3. Exit

2

Entries at index 0: No Hash Entry

Entries at index 1: 12 -> NULL

Entries at index 2: 3 -> 7 -> 15 -> NULL

Press 1. Insert 2. Display 3. Exit

3